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Cross Modal Point Cloud Completion with Symmetry Prior

対称性の事前知識を活用したクロスモーダル点群補完

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Abstract

Point cloud completion, a classic and enduring task in computer vision area, faces the persistent challenge of reconstructing high-fidelity, detail-rich point clouds with minimal computational cost [1]. Unlike traditional unimodal methods, view-guided point cloud completion (VIPC) proposed by [2] offers a promising alternative way by leveraging low-cost 2D images as supplementary information to enhance reconstruction detail. This article introduces a novel deep learning network for VIPC task by employing symmetry priors. Our method consists of a CSTNet, which learns geometric transformations and predicts a coarse point cloud skeleton of missing regions, and a Refine Module that subsequently upsamples and refines this skeleton. By effectively integrating the intrinsic geometric information of the point cloud with the supplementary structural cues provided by external 2D images, our method achieves high-quality global shape reconstruction and preserves fine-grained local details. Experimental results on the ShapeNet-VIPC benchmark [2] demonstrate that our method significantly outperforms existing state-of-the-art techniques.

1 Introduction

Point clouds, a widely adopted 3D data structure in industrial applications, are crucial for many industry technologies application [3]. However, real-world scanning devices such as LiDAR and depth cameras often capture sparse, unevenly distributed, and incomplete point clouds. [1]. To solve this, PCC(Point Cloud Completion) study is becoming more and more practical.

Traditional PCC methods predict missing parts based on mathematical methods, but the effect is relatively bad [1], driven by breakthroughs in artificial intelligence and deep learning. The current state-of-the-art approaches predominantly utilize deep learning models, with most method are fully supervised methods [4, 5, 6, 7]. Recognizing the difficulty of acquiring ground truth (GT) point clouds in practical situation, there's also a growing focus on developing self-supervised.[8] and weakly supervised methods [9]. Furthermore, inspired by the remarkable success of generative models, particularly diffusion models [10], some new model has great potential to become the next generation shape completion method [11, 12].

As the advancement of large language models and attention mechanisms [13] has propelled multimodality into a significant research trend. In the realm of 3D data, numerous methods are already exploring the combination of point clouds and images [14]. The View-Guided PCC (VIPC) task [2] is a relatively novel cross-modal point cloud completion task. Early approaches [2, 15] follows the previous PCC method [7], and use additional information from 2D modal for the refinement of Points. More recent advancements have focused on more complex information fusion strategies and usually no need to refine the points can still get good result, XMFNet [9] streamlined the cross-modal task through an information fusion approach. EGI-INet [16] drew inspiration from Vision Transformers (ViT) [17] and Meta-Transformer [18] concepts to represent both point clouds and 2D images as a common token form for later fusion. The most recent research, MAENet [19], built upon 2-stage coarse-refine method and feature pyramid fusion. Unfortunately while it improved the uniformity of point cloud distribution and currently holds the top performance, its significantly increased parameter count substantially limits computational efficiency.

Point clouds, as unstructured data, inherently contain rich 3D spatial information. This characteristic has widely used by many self-supervised methods [8]. In fact, point cloud completion can also be transfered to a spatial geometric transformation problem. This approach offers a distinct advantage over normal encoder-decoder architectures, as it bypasses the information loss often associated with encoding and decoding processes, leading to superior detail reconstruction in generated point clouds [20]. Consequently, recent research has begun to utilize these geometric priors to aid in shape completion. For instance, GTNet [21] is a network specifically designed for point cloud completion that employs spatial operations like symmetry, rotation, translation, and scaling to reconstruct missing parts. More recently, SymmCompletion [20], a new method, introduced the use of partial symmetry prior features for point cloud completion, enhancing the robustness of geometric transformation methods when dealing with asymmetric objects. This particular method served as a primary inspiration for our own approach.

Motivated from these facts, we hypothesize that the rich structural and semantic information in 2D images can also support the learning of more accurate geometric transformations for completion. To validate this idea, we propose a novel completion network which leverages image-guided symmetric priors.And Our method's unique contributions can be illustrated as following：

1. We are the first to convert cross-modal point cloud completion task to geometric transformations task.

2. Our model consists of two specialized modules: a transformation prediction network，CSTNet, it tries to learn to generate the missing point cloud by applying learned transformations to the partial input, and a refinement network that globally optimizes the completed shape for detail and uniformity.

3. Through experiments, we prove our algorithm realized the best performance on ShapeNet-VIPC dataset benchmark [2] while remaining computationally efficient.

2 Related work

2.1 3D Point Cloud Completion

Early PCC approaches established a dominant coarse-to-fine pipeline. Pioneering this field, PCN [7] adopts a two-stage framework: it initially produces a coarse, yet structurally complete, point cloud skeleton, which is subsequently refined by a decoder [22] to recover fine-grained details in the final output. This foundational design was adopted and enhanced by numerous subsequent methods [4, 23]. More recently, an alternative strategy has emerged that aims to preserve the original structure of the partial input. These methods, such as the Transformer-based Pointr [6], focus exclusively on predicting only the missing regions, which are then merged with the known points. While this approach effectively retains the input geometry, it faces a significant challenge: a mismatch in point density between the original and predicted regions often creates noticeable holes, gaps, and a non-uniform appearance, degrading the quality of the final shape [20].

2.2 VIPC task

As multimodal fusion is gaining more and more attention. As image is easy to get and can be a suitable additional information for point cloud analysis. In this situation, cross modal image-guided PCC task was introduced by VIPC [2], which also contributed the first benchmark dataset — ShapeNet-VIPC [2] for this problem. The first VIPC model follows the coarse-to-fine pipeline proposed by PCN [7], first using a Transformer for single-view reconstruction and then fusing this with the partial input before a image-support final refinement stage. Subsequent works try to use the cross-modal fusion strategy for this task. XMFNet [9] employs multiple layers of self-attention mechanisms and cross-attention mechanisms, tring to effectively fuse image and point modal features in the latent space, achieving strong results without a dedicated refinement module. EGIINet [16] proposes an explicit information interaction strategy, tokenizing both modalities to create a highly parameter-efficient model. More recently, MAENet [19] utilizes a multi-layer feature pyramid for fusion, followed by a refinement network to ensure a uniform point distribution. However, the refinement network will also introduce high computational cost.

2.3 Use of symmetry priors

Leveraging geometric priors is a powerful tool in 3D shape completion, with symmetry being a particularly effective cue [21]. This prior has been successfully applied before [8, 24]. In the PCC area, GTNet [21] successfully reconceptualized completion problem as a geometric transformation problem. Instead of directly generating points, it learns transformations (sucn as rotation, translation, scaling) to replicate existing parts of an object to fill in missing regions. Because these transformations are flexible and not limited to simple mirroring, this approach is also effective for asymmetric objects. Building on this, a recent work named SymmCompletion [20] proposed using local symmetry priors extracted from local features. This focus on local symmetry offers even greater applicability to complex, asymmetric shapes and serves as a key inspiration for our method. Our work aims to integrate these powerful symmetry-based transformation concepts into the view-guided completion framework.

3 Our Proposed Method

3.1 Overview of the whole Network

Our method addresses the view-guided completion task through a novel two-stage architecture. The core objective is to reconstruct a high-fidelity point cloud $P_{complete} \in \mathbb{R}^{N_q \times 3}$ from the input sparse partial point cloud $P_{in} \in \mathbb{R}^{N_p \times 3}$ single-view image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ as additional information for guidence. Our framework decouples this challenge into two distinct subproblems: coarse structure completion and image-guided refinement module. The overall pipeline can be checked in Fig. 1.

In the first stage, handled by our **Cross Symmetry transformation Network (CSTNet)**, focuses on generating a complete but coarse representation of the object's geometry. Instead of directly predicting the entire shape, CSTNet leverages local symmetry priors to infer the geometry of the missing regions. This predicted missing geometry P_m is then merged with a downsampled set of keypointsP_k to form a coarse point cloud, represented by$\mathbf{P}_{coarse}$.

Following second stage employs a **Refinement Module** to enhance the geometric detail and ensure point distribution uniformity of $\mathbf{P}_{coarse}$. Critically, this module leverages rich texture and structural features extracted from the additional input image $\mathbf{I}$ to guide the next refinement process, correcting inaccuracies and adding fine details that the previous CSTNet might miss.

Finally, to produce the output with a standardized point count, the refined point cloud followes a downsampling operation using Farthest Point Sampling (FPS), a very common downsampling method. This step ensures the final output $P_{complete}$ is a complete, detailed, and uniformly distributed point cloud with N_c points. The specific implementations of CSTNet and the Refinement Module are detailed in the subsequent sections.

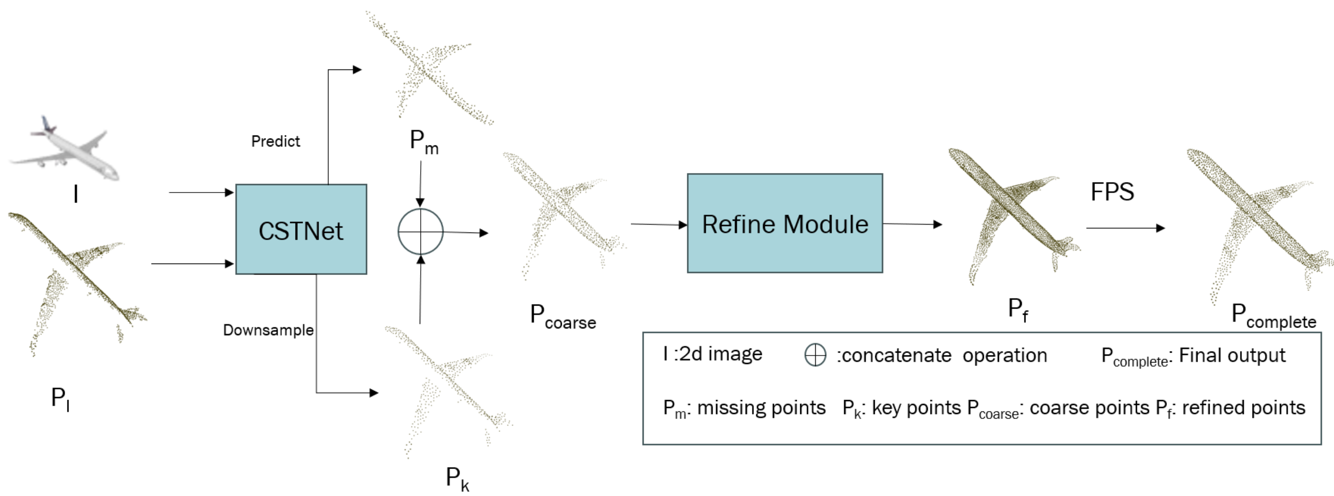

Fig. 1: An illustration of our proposed two-stage framework. By giving a uncomplete sparse point cloud and its correspond 2D image in the dataset, the model first employs the CSTNet for generating the initial coarse point cloud by predicting the missing geometry. Subsequently, followed Refinement Module enhances this coarse output, producing a detailed and uniform point cloud with a standard density (2048 points).

3.2 Cross Symmetry transformation Network (CSTNet)

The Cross Symmetry transformation Network (CSTNet), which is illustrated in Fig. 2, serves as the foundational module in our framework. Its primary function is to coarse point cloud which is nearly complete by learning an image-guided geometric transformation that exploits symmetry priors. The architecture of CSTNet can be broken down into three main stages: intra-modal encoding, cross-modal fusion, and transformation-based point generation.

1. Dual-Stream Intra-Modal Encoding We begin by extracting features from each modality in parallel.

- **Image Feature Extraction:** For the image stream, we employ a lightweight yet powerful ResNet18 [25] as the image encoder to extract a feature map $\mathbf{F}_\mathrm{I}$ containing rich semantic and textural information.
- **Point Cloud Feature Extraction:** For the geometry stream, we process the input point cloud $\mathbf{P}$ using a hierarchical downsampling architecture detailed in Fig. 3. This process, which utilizes Set Abstraction layers from PointNet++ [26] followed by a Point Transformer [27], yields two critical outputs: a set of downsampled keypoints $\mathbf{P}_k$ that form a sparse skeleton of the object, and their corresponding high-dimensional features $\mathbf{F}_k$.

2. Cross-Modal Feature Fusion To enrich the geometric features with visual context from the image, we fuse the features from both streams. Following recent successful approaches [16, 28, 9], we use a cross-attention mechanism to enable 2D and 3D information fusion between $\mathbf{F}_\mathrm{I}$ and $\mathbf{F}_k$. To enhance this interaction, the attention layers share parameters, a technique noted for its effectiveness in [16]. The fusion process is defined as:

$$\mathbf{F}'_\mathrm{I} = \gamma(\mathbf{F}_\mathrm{I}, \mathbf{F}_k) \tag{1}$$

$$\mathbf{F}'_k = \gamma(\mathbf{F}_k, \mathbf{F}_\mathrm{I}) \tag{2}$$

where γ represents the combined cross-attention and self-attention block, and $\mathbf{F}'_\mathrm{I}$ and $\mathbf{F}'_k$ are the resulting cross-modal features.

3. Coarse Point Generation Inspired by GTNet [21] and SymmCompletion [20], we formulate the prediction of the missing points as a spatial transformation problem. Instead of directly regressing point coordinates, we use the fused keypoints features $\mathbf{F}'_k$ to predict the parameters of an affine transformation. Two prediction heads, implemented as lightweight MLPs, are used for this purpose:

$$\mathbf{P}_m = \mathbf{P}_k \cdot \{\mathcal{X}(\mathbf{F}'_k)\} + \mathcal{Y}(\mathbf{F}'_k) \tag{3}$$

Here, $\mathcal{X}(\cdot)$ and $\mathcal{Y}(\cdot)$ are the MLP heads that predict the transformation matrix $\mathbf{A} \in \mathbb{R}^{N_k \times 3 \times 3}$ and the translation vector $\mathbf{T} \in \mathbb{R}^{N_k \times 3}$, respectively.

Finally, the coarse point cloud $\mathbf{P}_\mathrm{coarse}$ is formed by concatenating the original keypoints and the newly generated missing points: $\mathbf{P}_\mathrm{coarse} = \mathbf{P}_k \cup \mathbf{P}_m$. This complete but coarse shape is then passed to the refinement module.

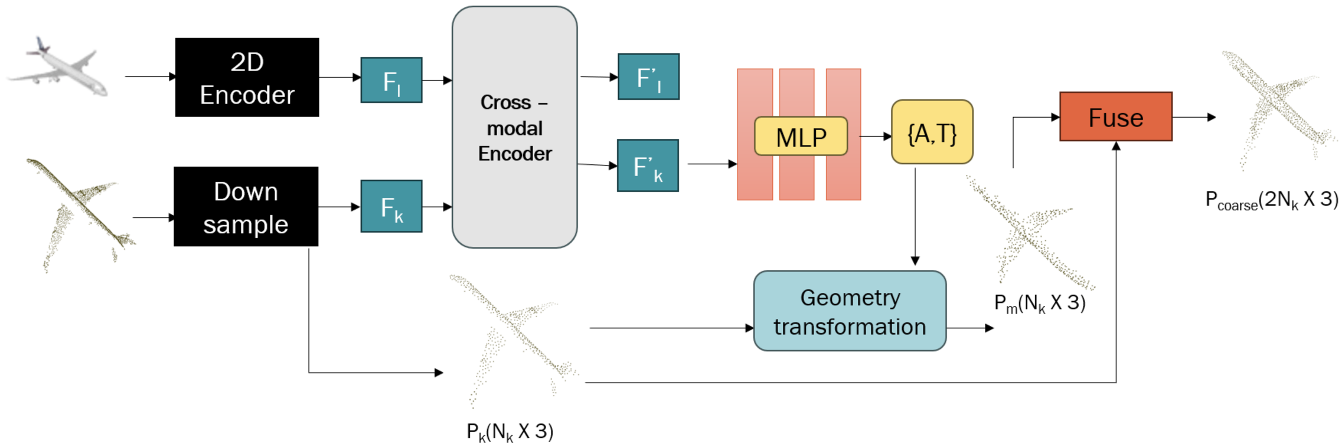

Fig. 2: The main architecture of our Cross Symmetry transformation Network (CSTNet). This network first extracts parallel features from the image and point cloud using intra-modal encoders. These features are then fused via a cross-modal attention mechanism. Finally, a prediction head (MLP) uses the fused features to regress the parameters of a geometric transformation, which is applied to input keypoints to generate the missing geometry.

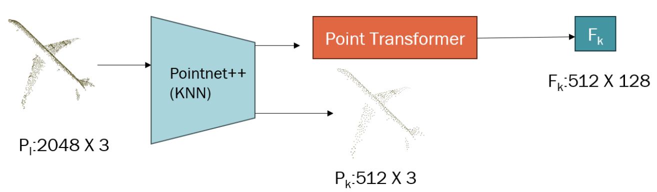

Fig. 3: Detailed architecture of the point cloud down-sampling module. This module employs PointNet++ style Set Abstraction layers followed by a Point Transformer to extract hierarchical features while producing a sparse set of keypoints.

3.3 Image-Guided Refinement Module

The coarse point cloud $\mathbf{P}_{\text{coarse}}$ generated by CSTNet provides a complete geometric structure but may lack fine details and uniform point distribution. The purpose of the Refinement Module, inspired by recent advances in the field [20, 29], is to address these issues by leveraging a rich set of multi-source information. As depicted in Fig. 4, this module takes four distinct feature sets as input: the coarse point features ($\mathbf{F}_{\text{coarse}}$), the keypoint features ($\mathbf{F}_k$), the predicted missing point features ($\mathbf{F}_m$), and the global image features ($\mathbf{F}_{\text{I}}$). How to effectively fuse features from so many different sources is one of the main challenges of this module.

1. Multi-Source Information Fusion The core of this module is a hierarchical fusion pipeline designed to inject both visual and geometric guidance into the coarse features.

- **Image Cross-modal Guidence:** To leverage the 2D image, we first fuse the image 2D modal features $\mathbf{F}_{\text{I}}$ with both the coarse initial point cloud features $\mathbf{F}_{\text{coarse}}$ and the predicted missing part points features $\mathbf{F}_m$. This can be achieved by using a cross-attention mechanism where 2D image serves as the context (key/value) for the point features (query), thereby enriching the geometric representations with detailed visual cues from the image.
- **Geometric Self-Guidance:** Next, to ensure the refined shape respects the original known geometry and the predicted structure, referring to [20]'s method, we use the "parts" to guide the "whole". Specifically, we do fusion operation to the fused coarse features $\mathbf{F'}_{\text{coarse}}$ with both the fused keypoint features $\mathbf{F'}_k$ and the fused missing point features $\mathbf{F'}_m$. This allows the high-confidence information from the original partial scan ($\mathbf{F}_k$) and the structural information from the predicted part ($\mathbf{F}_m$) to guide the refinement of the overall shape.

This multi-level fusion process, employing combined cross-attention and self-attention blocks (η, θ), culminates in a highly informed feature representation, $\mathbf{F}''_{\text{coarse}}$, for the coarse point cloud:

$$\mathbf{F}''_{\text{coarse}} = \left[\eta(\mathbf{F}'_{\text{coarse}}, \mathbf{F}'_k)\,;\,\theta(\mathbf{F}'_{\text{coarse}}, \mathbf{F}'_m)\right] \tag{4}$$

where $\mathbf{F}'$ denotes features that have been enriched by the image context, and $[\cdot\,;\,\cdot]$ represents feature concatenation. The detailed architecture of our fusion layer design can be checked in Fig. 5.

2. Decoder and Detail Enhancement The final stage translates the enhanced features into a high-fidelity point cloud. First, the fused feature $\mathbf{F}''_{\text{coarse}}$ is passed through two consecutive self-attention layers, denoted by $\zeta(\cdot)$, for final feature propagation and enhancement.

Instead of a standard deconvolutional decoder, we adopt an efficient upsampling strategy based on learned point displacements. We first duplicate the points in $\mathbf{P}_{\text{coarse}}$ and then predict a 3D offset for each point using a lightweight MLP that takes the enhanced features as input. This process is formulated as:

$$\mathbf{P}\text{f} = \mathbf{P}\text{up} + \Delta\mathbf{P} \tag{5}$$

where

$$\mathbf{P}\text{up} = \mathcal{U}(\mathbf{P}\text{coarse}) \quad \text{and} \quad \Delta\mathbf{P} = \text{MLP}(\zeta(\mathbf{F}''\text{coarse})) \tag{6}$$

$\mathbf{P}$up denote the upsampled point cloud. $\Delta\mathbf{P}$ is the offset of every points.

To conclude, and in line with methods like MAENet [19], this dense cloud is downsampled to the standard 2048 points. This upsampling-then-downsampling strategy has been shown to be effective at producing a final point cloud with a highly uniform distribution.

3.4 Loss Function

Loss function is very important in our training process, the training of our network is guided by a composite loss function, $\mathcal{L}_{\text{total}}$, which is designed to enforce both geometric fidelity in the reconstructed point cloud and meaningful fusion between the 2D and 3D modalities. Specifically it consists of two components: Point Completion loss and cross-modal style transfer loss.

Point Completion Loss. To ensure the geometric accuracy of the output, we employ the Chamfer distance implemented by L1 norm regularization (CD_{l1}), a classic and wide-used metric. It can be defined as follow:

$$\begin{aligned} d_{\text{CD}}(P_1, P_2) = \frac{1}{|P_1|} \sum_{n_1 \in P_1} \min_{n_2 \in P_2} \|n_1 - n_2\|_2 + \\ \frac{1}{|P_2|} \sum_{n_2 \in P_2} \min_{n_1 \in P_1} \|n_2 - n_1\|_2 \end{aligned} \tag{7}$$

Here, P_1, P_2 stands for the two sets of points to be compared. In order to give full utilize the role of each model stage, a key aspect of our training strategy is application of multi-stage supervision. Instead of only supervising the final output, we apply the CD loss at multiple stages of our pipeline to ensure accuracy is maintained throughout the generation process. Our total completion loss, $\mathcal{L}_{\text{CD}}$, is the sum of these individual losses:

$$\mathcal{L}_{\text{CD}} = d_{\text{CD}}(\mathbf{P}_{\text{coarse}}, \mathbf{Q}) + d_{\text{CD}}(\mathbf{P}_{\text{f}}, \mathbf{Q}) + d_{\text{CD}}(\mathbf{P}_{\text{complete}}, \mathbf{Q}) \tag{8}$$

$\mathbf{Q}$ is the ground truth point cloud, as the terms $\mathbf{P}_{\text{coarse}}$, $\mathbf{P}_{\text{f}}$, $\mathbf{P}_{\text{complete}}$ respectively represent the outputs from the CSTNet, the Refinement Module, and the final downsampling step, respectively. This approach provides direct gradients to each major component of our network.

Style Transfer Loss. To encourage effective feature fusion between the 2d image and 3d point cloud modalities, we incorporate a style transfer loss, which has proven effective in recent cross-modal methods [16, 28]. This loss enforces consistency in the second-order statistics (i.e., style) of the feature distributions before and after the cross-modal interaction and thus enhance the quality of information fusion. It is formulated as:

$$\mathcal{L}_{\text{style}} = \frac{\left\|G(\mathbf{F}_i) - G(\mathbf{F}'_p)\right\|_F^2 + \left\|G(\mathbf{F}_p) - G(\mathbf{F}'_i)\right\|_F^2}{n_p \times C} \tag{9}$$

Here, $\mathbf{F}_p$ and $\mathbf{F}_i$ are the intra-modal features in the CSTNet process, while $\mathbf{F}'_p$ and $\mathbf{F}'_i$ are the features after cross-modal fusion. $G(\cdot)$ is the Gram matrix function, defined as $G(\mathbf{F}) = \mathbf{F}^T\mathbf{F}$, and $\|\cdot\|_F^2$ denotes the squared Frobenius norm.

Overall Loss Our overall loss function is constructed by aggregating the point completion loss and the style transfer losses, with appropriate weighting to balance their contributions.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CD}} + \alpha \cdot \mathcal{L}_{\text{style}} \tag{10}$$

where α is a hyperparameter that balances the contribution of each term. In our implementation, we empirically set α to a fixed value of 0.03.

4 Experiments

4.1 Experimental Settings

Dataset. The entirety of our experimental evaluation is carried out on the **ShapeNet-VIPC** dataset [2].It contains 8 types of objects for normal test and 5 unseen types of objects for generation ability test. All input settings and evaluation metrics remain exactly the same as previous methods. Each object in the dataset is associated with 24 partial point clouds, each generated from a unique viewpoint and paired with a corresponding 2D image, simulating real-world occlusion scenarios. To realize a fair and direct comparison with prior method, we adhere strictly to the data splits method and experimental protocols used in [2, 15, 9, 16, 28, 19].

Evaluation Metrics. We evaluate the ability of our current model by using two standard and wide used metrics: L2 norm regularized Chamfer distance (l_2-CD) [30] together with F-Score [31]. The two matrics are formulated in formulation 11 and 12.

$$\begin{aligned} d_{\text{CD}}(P_1, P_2) = \frac{1}{|P_1|} \sum_{n_1 \in P_1} \min_{n_2 \in P_2} \|n_1 - n_2\|_2^2 \\ + \frac{1}{|P_2|} \sum_{n_2 \in P_2} \min_{n_1 \in P_1} \|n_2 - n_1\|_2^2 \end{aligned} \tag{11}$$

$$F\text{-Score}(i) = \frac{2 \cdot \text{Pre}(i) \cdot \text{Re}(i)}{\text{Pre}(i) + \text{Re}(i)} \tag{12}$$

Here Pre means precision, Re means recall.

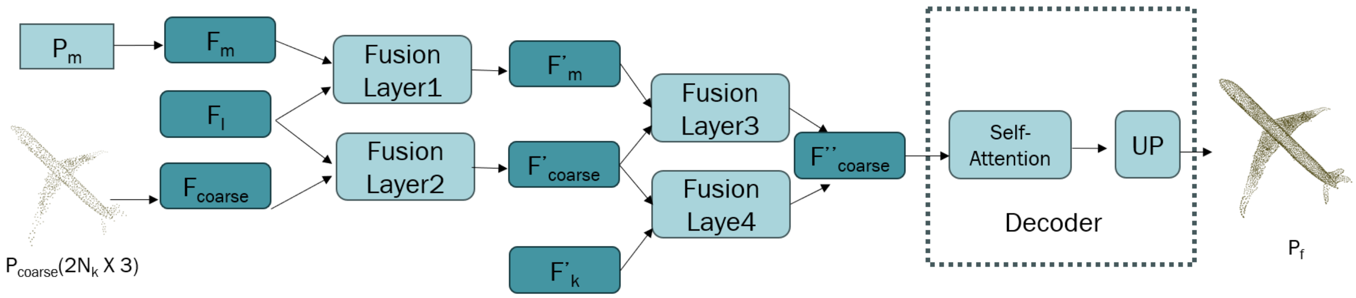

Fig. 4: Overview of the Refinement Module architecture. The module first processes the coarse point cloud and predicted missing points using parallel Local-Encoders to extract distinct feature sets. These features, along with image and keypoint features, are then progressively fused through a series of identical Fusion Layers to produce the final refined shape.

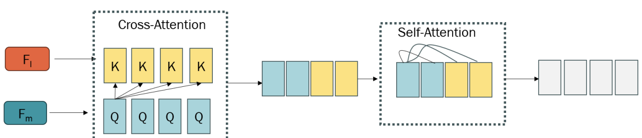

Fig. 5: Detailed structure of a single Fusion Layer. This layer combines a cross-attention block for inter-modal information exchange (e.g., image features as key/value, point features as query) with a self-attention block for intra-modal feature enhancement.

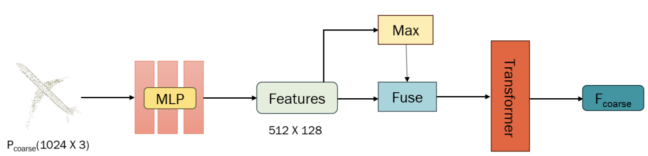

Fig. 6: The detailed architecture of the Local Encoder. It processes a point set by first using MLPs for point-wise feature extraction, followed by a max-pooling operation and a Transformer block for contextual feature aggregation.

4.2 Implementation Details

The input point cloud has been initially preprocessed to a standard size, 2048 points via Farthest Point Sampling (FPS),image is 224×224 low resolution RGB. Final network output is our predict complete point cloud, it also contains 2048 points, matching the ground truth density.

Within our architecture, the CSTNet module first downsamples the input point cloud to 512 keypoints. It then generates an additional 512 points, forming a 1024-point coarse cloud. The Refinement Module subsequently upsamples this coarse cloud by a factor of 4 to a dense 4096-point cloud, which is finally downsampled via FPS to the standard 2048-point-resolution output.

All models were developed and trained by using PyTorch and CUDA. Batch size is set 64 for 200 training epochs.

4.3 Results and Analysis

Quantitative Comparison. In our experiment, We conducted a comprehensive test and evaluation of our proposed model on eight representative categories from the Shapenet-VIPC dataset [2]. Our method was benchmarked against a suite of latest approaches.

The quantitative results, which are presented in Tab 1 and Tab 2, show the performance superiority of our network. Our model realizes the lowest l_2-Chamfer Distance across all of the categories except for the "couch". Similarly, the F-Score results confirm that our method produces completions that have a better overlap with the ground truth geometry compared to all other competing methods except the same one. However, the gap between us and the current best method MAENet [19] is not that big.

Model Size and Efficiency. In addition to quantitative performance, we also analyze the parameter efficiency of our model. Table 3 presents a comparison of model sizes against recent leading methods. However, our refinement module introduces additional parameters, therefore, our method does not have a leading advantage in terms of parameter usage

Notably, our framework strikes a favorable balance between performance and efficiency. It significantly outperforms the previous best-performing model, MAENet [19], while maintaining significantly smaller parameter count. Future work will focus on architectural optimizations without influencing the completion quality.

Table 1: Mean Chamfer Distance per point (CD $\times 10^3 \downarrow$).

Methods	Airplane	Cabinet	Car	Chair	Lamp	Sofa	Table	Watercraft
PCN [7]	4.246	6.409	4.840	7.441	6.331	5.668	6.508	3.510
PoinTr [6]	1.686	4.001	3.203	3.111	2.928	3.507	2.845	1.737
PointAttN [5]	1.613	3.969	3.257	3.157	3.058	3.406	2.787	1.872
ViPC [2]	1.760	4.558	3.183	2.476	2.867	4.481	4.990	2.197
CSDN [15]	1.251	3.670	2.977	2.835	2.554	3.240	2.575	1.742
XMFNet [9]	0.572	1.980	1.754	1.403	1.810	1.702	1.386	0.945
EGIINet [16]	0.534	1.921	1.655	1.204	0.776	1.552	1.227	0.802
MambaTron [28]	0.537	1.905	1.620	1.184	0.774	1.561	1.234	0.779
MAENet [19]	0.483	1.608	1.524	1.053	0.624	**1.317**	1.034	0.687
Ours	**0.446**	**1.589**	**1.499**	**1.039**	**0.562**	1.342	**1.026**	**0.660**

Table 2: Mean F-Score @ 0.001 $\uparrow$.

Methods	Airplane	Cabinet	Car	Chair	Lamp	Sofa	Table	Watercraft
PCN [7]	0.578	0.270	0.331	0.323	0.456	0.293	0.431	0.577
PoinTr [6]	0.842	0.516	0.545	0.662	0.742	0.547	0.723	0.780
PointAttN [5]	0.841	0.483	0.515	0.683	0.729	0.512	0.699	0.774
ViPC [2]	0.803	0.451	0.512	0.529	0.706	0.434	0.594	0.730
CSDN [15]	1.251	3.670	2.977	2.835	2.554	3.240	2.575	1.742
XMFNet [9]	0.961	0.662	0.691	0.809	0.792	0.723	0.830	0.901
EGIINet [16]	0.969	0.693	0.723	0.847	0.919	0.756	0.857	0.927
MambaTron [28]	0.969	0.697	0.728	0.851	0.920	0.764	0.856	0.931
MAENet [19]	0.981	0.754	0.756	0.885	0.945	**0.818**	0.895	0.951
Ours	**0.988**	**0.756**	**0.762**	**0.886**	**0.953**	0.806	**0.897**	**0.956**

Table 3: Comparisons on model sizes

Methods	Number of parameters (M)
ViPC [2]	17.43
CSDN [15]	16.85
XMFnet [9]	9.57
EGIINet [16]	**9.03**
MAENet [19]	32.72
Ours	18.80

5 Conclusion and Future Work

We present a novel network designed to address the cross-modal point cloud completion task (VIPC task) that effectively leverages geometric symmetry priors in this paper. Our network is a two-stage architecture which firstly generates a coarse structure by learning image-guided geometric transformations, and then progressively refines it to produce a high-fidelity, uniformly distributed output. The Experiments validate our approach and validate that our model performs best on the Shapenet-VIPC benchmark.

Building upon this successful foundation, we identify two primary directions for future research. First, we will focus on **architectural optimization**. The current model, while effective, contains numerous components. We plan to conduct thorough ablation studies and explore techniques such as model pruning to streamline the network, creating a more lightweight yet equally powerful version. Second, we aim to investigate the model's **generalization capabilities**. Future work will also involve evaluating the ability of our approach on diverse real-world datasets, such as the large-scale outdoor scenes in the KITTI benchmark [32].

References

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